

CSE252

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

LTP: 2-0-2 Cr: 3

Lecture #0

A kick start session



Course Outcomes

CO1 ::

Understand the fundamental concepts of Artificial Intelligence, Machine Learning, intelligent agents, and their applications in robotics and autonomous systems.

CO2 ::

Apply preprocessing, feature engineering, and visualization techniques on datasets for developing AI models.

CO3 ::

Build supervised machine learning models for solving regression and classification problems using appropriate algorithm.

CO4 ::

Apply unsupervised learning techniques for clustering and dimensionality reduction to discover hidden patterns in data.

CO5 ::

Examine and improve machine learning models using appropriate performance metrics and validation techniques.

CO6 ::

Build simple AI-enabled solutions for real-world robotics and automation problems using Python and Scikit-learn.

School Vision

To be a globally recognized school through excellence in teaching, learning and research for creating Computer Science professionals, leaders and entrepreneurs of future contributing to society and industry for sustainable growth.

School Mission

- To build computational skills through hands-on and practice-based learning with measurable outcomes.
- To establish a strong connect with industry for in-demand technology driven curriculum.
- To build the infrastructure for meaningful research around societal problems.
- To nurture future leaders through research-infused education and lifelong learning.
- To create smart and ethical professionals and entrepreneurs who are recognized globally.

Program Outcomes

PO1: Engineering knowledge

- Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis

- Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions

- Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems

- Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage

- Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society

- Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

Program Outcomes (*Contd.*)

PO7: Environment and sustainability

- Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics

- Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work

- Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication

- Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance

- Demonstrate knowledge and understanding of the engineering, management principles and apply the same to one's own work, as a member or a leader in a team, manage projects efficiently in respective disciplines and multidisciplinary environments after consideration of economic and financial factors.

PO12: Life-long learning

- Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes

PSO1

- Apply acquired skills in software engineering, networking, security, databases, intelligent systems, cloud computing and operating systems to adapt and deploy innovative software solutions for diverse applications.

PSO2

- Apply diverse IT skills to design, develop, and evaluate innovative solutions for business environments, considering risks, and utilizing interdisciplinary knowledge for efficient real-time projects benefiting society.



CO-PO and CO-PSO Articulation Matrix

PO & PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	-	-	2	-	1	-	-	-	2	3	1
CO2	3	3	2	2	3	-	-	-	-	-	-	1	2	2
CO3	3	3	3	2	3	-	-	-	-	-	-	1	3	2
CO4	3	3	2	3	3	-	-	-	-	-	-	1	3	2
CO5	2	3	2	3	3	-	-	-	-	-	-	1	2	2
CO6	3	2	3	2	3	2	-	-	2	1	1	2	3	3

Legend:

The mapping uses the standard accreditation correlation levels:

3: Substantial (High)

2: Moderate (Medium)

1: Slight (Low)

Course Focus: Orientation and Cohort Mapping

Orientation

Knowledge Enhancement

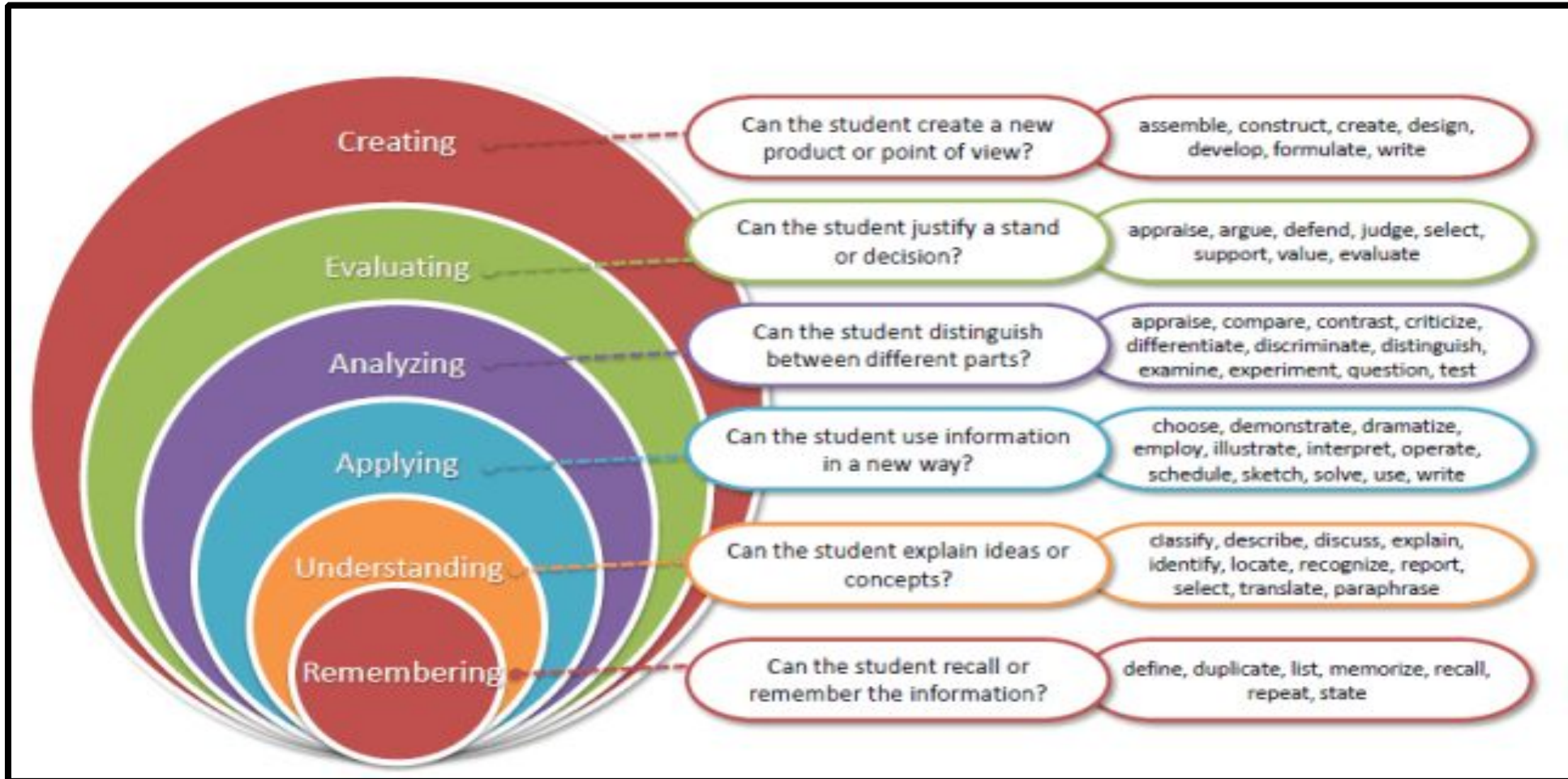
Placement Examination

Skill Enhancement

Cohort

Cohort-5:
Machine Learning

Revised Bloom's Taxonomy



Why This Course?

The Intelligence Taxonomy

Artificial Intelligence	Machine Learning (Course Focus)	Deep Learning	Data Science
Scope: Broadest	Scope: AI Subfield	Scope: ML Subfield	Scope: Cross-disciplinary
Core Approach: Mimics human logic	Core Approach: Learns from data	Core Approach: Neural networks	Core Approach: Uses ML as a tool
Human Intervention: High (Rule-based)	Human Intervention: Moderate (Feature Engineering)	Human Intervention: Low	Human Intervention: High (Human-led analysis)
Output/Goal: Optimal action	Output/Goal: Accurate prediction	Output/Goal: Complex pattern extraction	Output/Goal: Business insight

Why This Course?

THE ECOSYSTEM: DEFINING THE BOUNDARIES OF INTELLIGENCE

Data Science

The overarching discipline of extracting insights from data using statistical methods.

Artificial Intelligence (AI)

Systems designed to mimic human intelligence, encompassing logic and problem-solving.

DATA SCIENCE

ARTIFICIAL INTELLIGENCE (AI)

MACHINE LEARNING (ML)

DEEP LEARNING

Machine Learning (ML)

Algorithms that learn and optimise patterns from data without explicit programming.

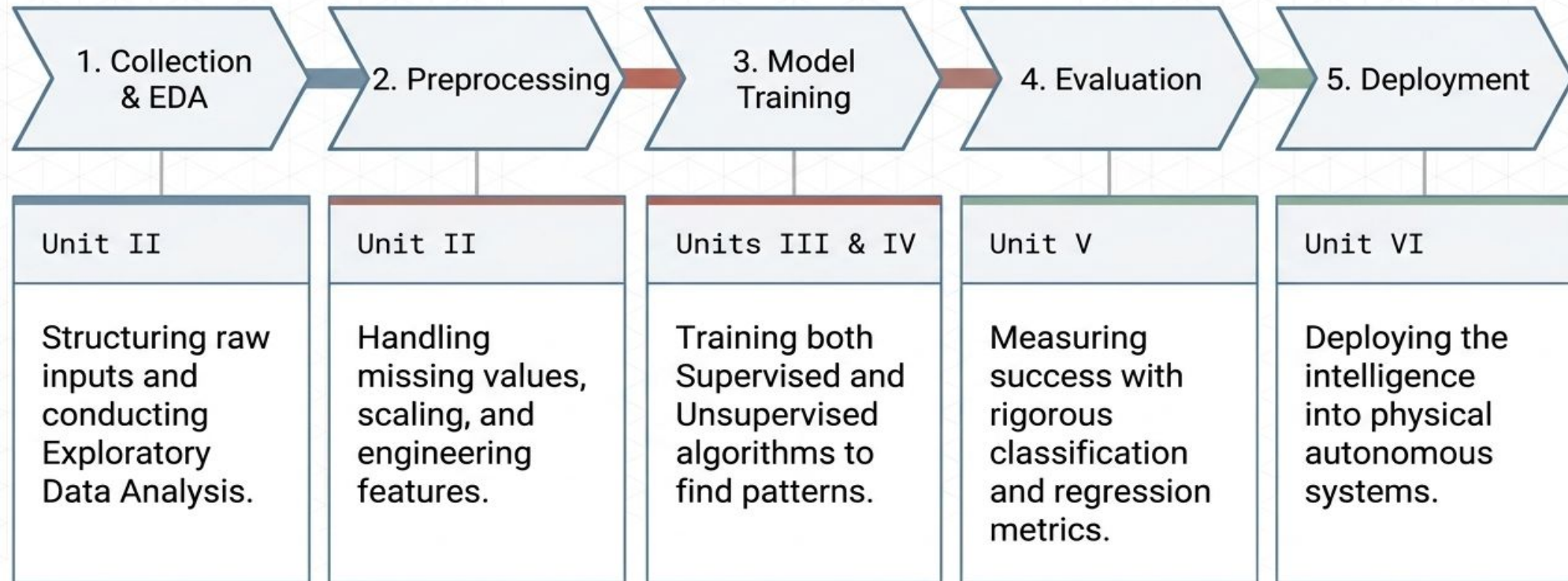
Deep Learning

A specialised subset using multi-layered neural networks for complex pattern recognition.

The course integrates all four to map the complete AI Development Lifecycle.

Why This Course?

The Machine Learning Pipeline



Why This Course?

1. Collection & EDA

2. Preprocessing

3. Model Training

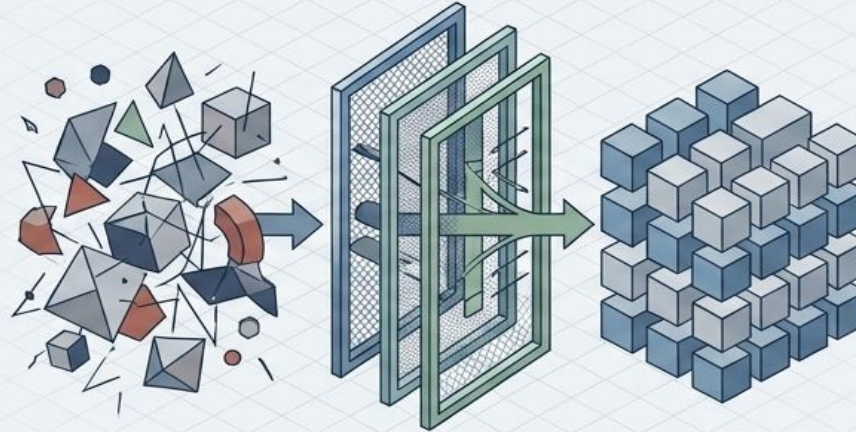
4. Evaluation

5. Deployment

Stage 1: Data Preparation & Engineering

The Challenge (Raw Inputs)

- Handling missing values
- Detecting statistical outliers
- Managing disparate data types



The Solution (Feature Engineering)

- Feature scaling for uniformity
- Feature encoding for categoricals
- Feature selection for relevance

Real-World Context: Students execute Exploratory Data Analysis (EDA) on raw inputs, including the Titanic dataset and Student Performance datasets.

Why This Course?

1. Collection & EDA

2. Preprocessing

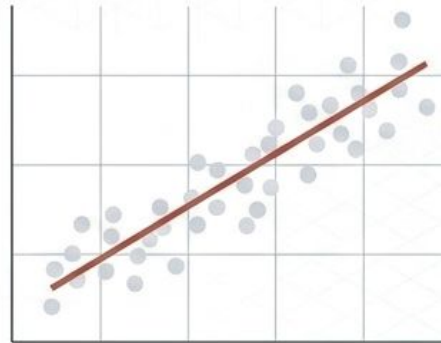
3. Model Training

4. Evaluation

5. Deployment

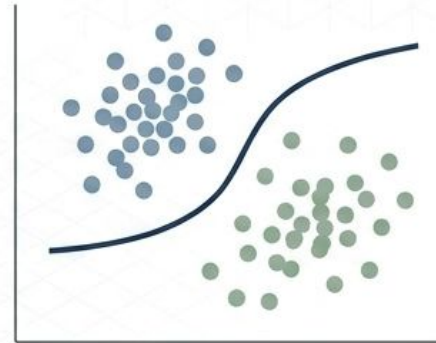
Stage 2: Supervised Learning (Learning with Labels)

Regression
(Predicting Continuous Values)



- Linear Regression
- Polynomial Regression
- Decision Tree Regression

Classification
(Predicting Distinct Categories)



- Logistic Regression
- K-Nearest Neighbour (KNN)
- Decision Tree & Random Forest
- Naïve Bayes

Why This Course?

1. Collection & EDA

2. Preprocessing

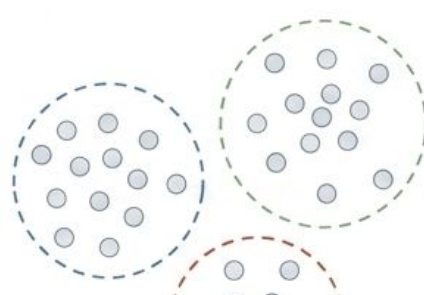
3. Model Training

4. Evaluation

5. Deployment

Stage 3: Unsupervised Learning (Discovering Patterns)

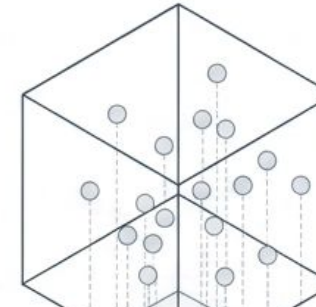
Clustering (Finding Hidden Groups)



Robotics Application: Pattern Discovery and clustering are essential for processing raw, unstructured sensor data in autonomous environments where labelled data is unavailable.

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN

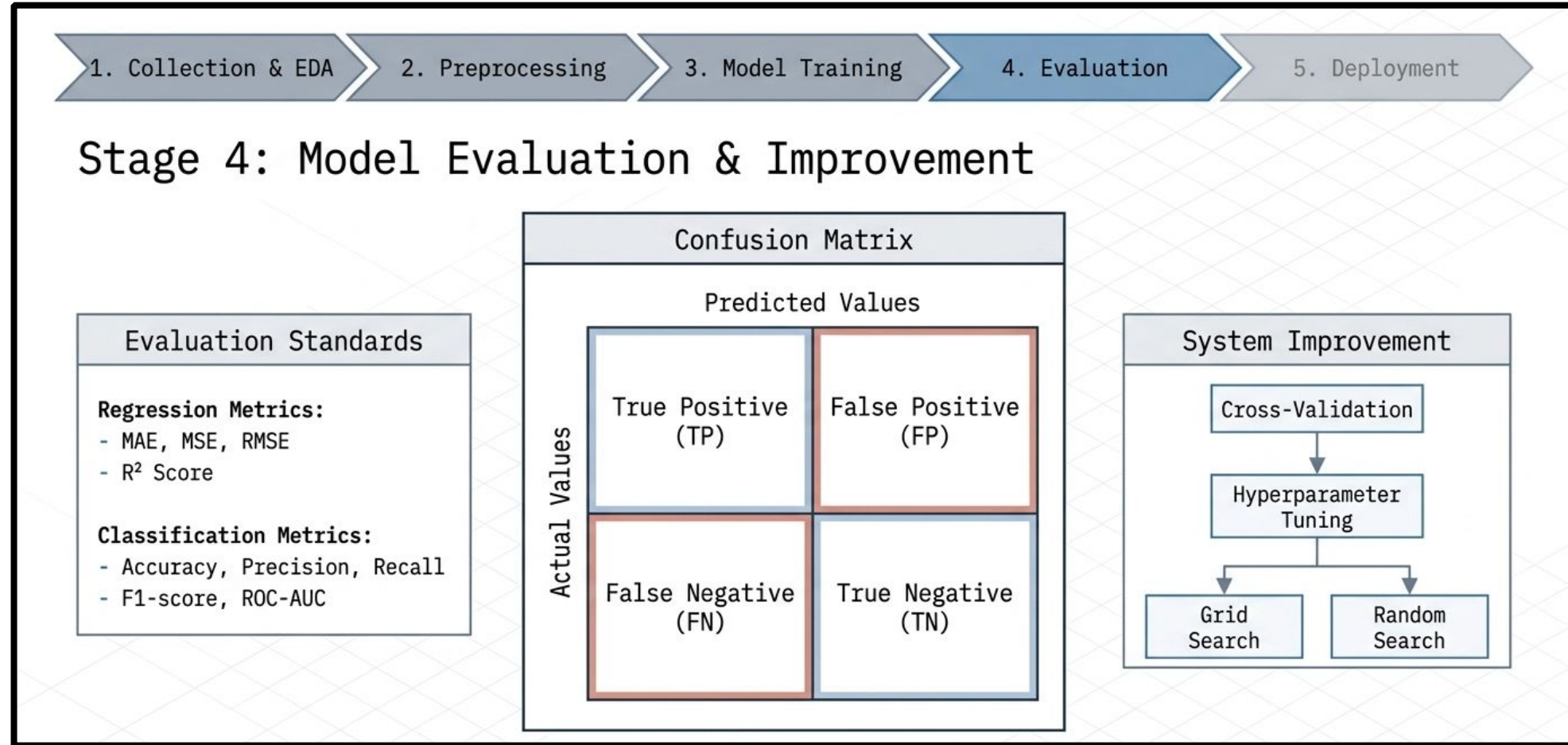
Dimensionality Reduction



- Principal Component Analysis (PCA)
- Association Rule Mining
- Apriori Algorithm

Robotics Application: Pattern Discovery and Clustering are essential for processing raw, unstructured sensor data in autonomous environments where labelled data is unavailable.

Why This Course?



Course includes an introduction to Explainable AI to ensure transparent and responsible model behavior prior to deployment.

Why This Course?

1. Collection & EDA

2. Preprocessing

3. Model Training

4. Evaluation

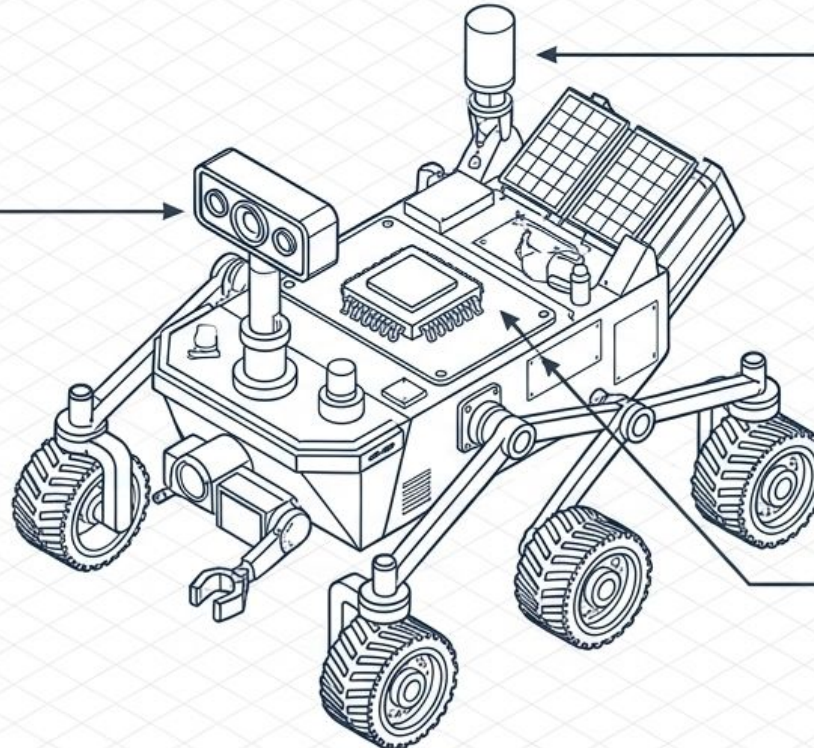
5. Deployment

The Capstone Synthesis: Intelligent Systems

Unit- 6

The Eyes: Computer Vision

Image Classification and Object Detection for real-time environmental awareness.



The Voice: Natural Language Processing

NLP Fundamentals, Chatbots, and Conversational AI for human-machine interaction.

The Brain: Reinforcement Learning

Agent, State, Action, and Reward loops driving autonomous decision-making.

Course Contents

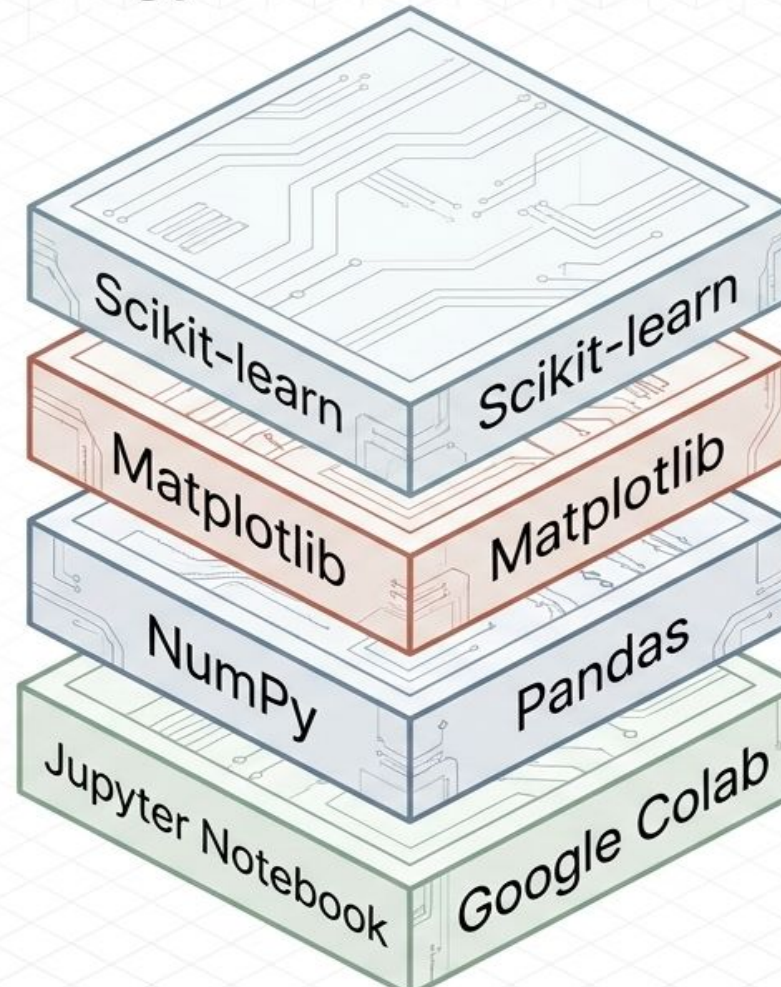
The Practical Technology Stack

Layer 4: Implementation

Layer 3: Visualisation

Layer 2: Data Manipulation

Layer 1: Environment



Industry-standard machine learning library for model deployment.

Charting and exploratory data analysis.

High-performance array processing and dataframe manipulation.

Rapid prototyping and cloud execution.

Course Syllabus

Before
MTE

UNIT-I

- **Foundations of Artificial Intelligence:** Introduction to AI and its evolution, Types of Artificial Intelligence, AI vs ML vs Deep Learning vs Data Science, Problem-solving using AI and AI Development Lifecycle, Intelligent Agents and Agent Environment, Applications of AI in Robotics, Healthcare, Manufacturing and Smart Cities, Ethics in AI and Responsible AI, Introduction to Generative AI

UNIT-II

- **Data Preparation and Machine Learning Workflow:** Introduction to NumPy, Pandas and Matplotlib, Understanding Data (Types of Datasets, Data Collection and Exploratory Data Analysis), Data Preprocessing (Handling missing values and Detecting outliers), Feature Engineering (Introduction to feature scaling, feature encoding and feature selection), Training and Validation (Train-Test Split and Cross Validation)

UNIT-III

- **Supervised Machine Learning:** Regression (Linear Regression, Polynomial Regression, Decision Tree Regression), Classification (Logistic Regression, K-Nearest Neighbour, Introduction to Decision Tree Classifier and Random Forest, Naïve Bayes), Concepts of Bias, Variance, Underfitting and Overfitting

Course Syllabus

After
MTE

UNIT-IV

- **Unsupervised Learning:** Clustering (K-Means Clustering, Hierarchical Clustering, Concept of DBSCAN), Dimensionality Reduction and Principal Component Analysis, Pattern Discovery (Association Rule Mining, Apriori Algorithm), Unsupervised Learning application in Robotics and Sensor data

UNIT-V

- **Model Evaluation and Improvement:** Regression Metrics (MAE, MSE, RMSE, R^2 Score), Confusion Matrix and Classification Metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC), Model Improvement (Cross Validation, Hyperparameter Tuning, Grid Search, Random Search), Introduction to Explainable AI

UNIT-VI

- **AI for Intelligent Systems and Robotics:** Computer Vision (Overview of Image Classification, Overview of Object Detection) for Intelligent Systems and Robotics, Natural Language Processing (NLP Fundamentals, Chatbots and Conversational AI) for Intelligent Systems and Robotics, Reinforcement Learning (Concepts - Agent, State, Action, Reward) in context Intelligent Systems and Robotics, AI for Autonomous Robots and Future Trends in AI for Intelligent Systems and Robotics

Course Syllabus | List of Practicals

Installation and familiarization with Python, Jupyter Notebook and Google Colab.

Data manipulation using NumPy and Pandas.

Data visualization using Matplotlib.

Exploratory Data Analysis on a real-world dataset – Titanic dataset and Student Performance dataset.

Data preprocessing: handling missing values, encoding and feature scaling.

Implementation of Linear Regression & Logistic Regression.

Classification using K-Nearest Neighbour, Decision Tree and Random Forest.

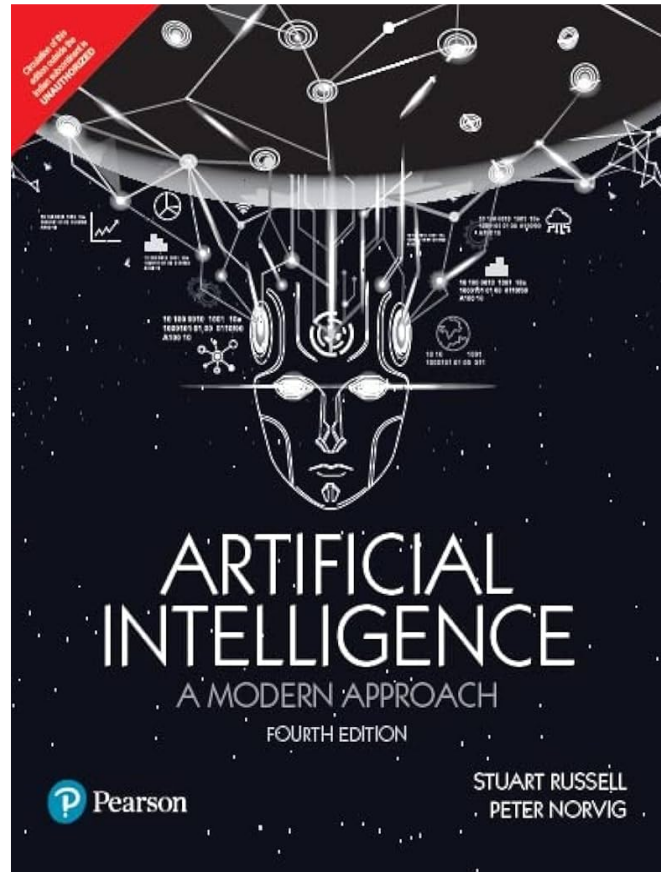
Clustering using K-Means.

Dimensionality Reduction using PCA.

Performance evaluation using Confusion Matrix, Precision, Recall and F1-score.

Hyperparameter tuning using Grid Search.

Mini Project: Development of an AI solution for a robotics, automation or smart system application.



Text Book:

ARTIFICIAL INTELLIGENCE: A MODERN APPROACH

By: STUART RUSSELL, PETER NORVIG

Publisher: PEARSON

Reference Books:

- A FIRST COURSE IN ARTIFICIAL INTELLIGENCE by Deepak Khemani, Mc Graw Hill
- DATA SCIENCE AND MACHINE LEARNING USING PYTHON by Reema Thareja, Mc Graw Hill
- PYTHON MACHINE LEARNING BY EXAMPLE: UNLOCK MACHINE LEARNING BEST PRACTICES WITH REAL-WORLD USE CASES by Yuxi (Hayden) Liu, Packt Publishing
- FUNDAMENTALS OF MACHINE LEARNING: A BUSINESS PERSPECTIVE by Pratyush Banerjee, Supriti Mishra, Shivashankar Chari, Mc Graw Hill
- MACHINE LEARNING WITH PYTORCH AND SCIKIT-LEARN: DEVELOP MACHINE LEARNING AND DEEP LEARNING MODELS WITH PYTHON by Sebastian Raschka, Yuxi (Hayden) Liu, Vahid Mirjalili, Packt Publishing

Pedagogical Initiatives

Detailed mapping of the pedagogical tools and activities to their respective lectures and topics, as outlined in the IP

Pedagogical Innovation	Description & Elaboration	Application Area (Specific Topics & Lectures)
Use of AI	Utilizing Generative AI tools (like ChatGPT or Gemini) for live demonstrations to explain concepts and explore prompt engineering. AI is also used to demonstrate model flaws by attempting to seek answers for the latest questions, highlighting AI ethics and limitations.	• <i>Lecture 4 & 5:</i> Applications in Smart Cities, Ethics in AI, and Introduction to Generative AI. • <i>Practical 14:</i> Mini-project development evaluating end-to-end AI solutions.
Case-Based Learning	Analyzing real-world scenarios to understand theoretical concepts, allowing students to map abstract AI principles to tangible industry problems.	• <i>Lecture 3:</i> Intelligent Agents (Automated vacuum cleaners/drones). • <i>Lecture 18:</i> Diagnostic case study emphasizing false negatives (Disease prediction models).

Pedagogical Initiatives

Detailed mapping of the pedagogical tools and activities to their respective lectures and topics, as outlined in the IP

Pedagogical Innovation	Description & Elaboration	Application Area (Specific Topics & Lectures)
Quizzes & Gamified Assessment	Implementing "Guess the Output" coding quizzes and gamified concept checks to assess understanding of syntax and logic on the fly.	• <i>Lecture 5</i>: Data Preparation (Introduction to NumPy, Pandas, and Matplotlib).
Flipped Classroom	Assigning foundational theory and specific video lectures as pre-class viewing. Class time is strictly reserved for higher-order discussions, complex problem-solving, and concept mapping.	• <i>Lecture 5</i>: Introduction to Python/Pandas (Utilizing AV-7: OpenStax Data Science).
Multimedia Tools	Leveraging 3D/2D visual projections, data distribution plots, and live webcam demonstrations to make abstract mathematical operations observable and intuitive.	• <i>Lecture 7</i>: Feature scaling (Visual before/after distribution plots). • <i>Lecture 15</i>: PCA (3D to 2D projection demonstration). • <i>Lecture 25</i>: Computer Vision (Live webcam bounding-box demonstration).

Pedagogical Initiatives

Detailed mapping of the pedagogical tools and activities to their respective lectures and topics, as outlined in the IP

Pedagogical Innovation	Description & Elaboration	Application Area (Specific Topics & Lectures)
Interactive Simulations & Role Play (Custom)	Employing physical role-playing exercises and interactive simulations to demonstrate algorithmic behavior (like moving centroids or reward loops) dynamically.	• <i>Lecture 13 & 14:</i> K-Means Clustering (Role play to team up students into 4 groups and choose a leader). • <i>Lecture 27:</i> Reinforcement Learning (Role play delivering a notebook from the first to the last seat to simulate Agent/Reward states).
Live-Coding Demonstrations (Custom)	Writing and executing Python/Scikit-learn code live in front of the class to demonstrate error handling, data preprocessing, and model building in real-time.	• <i>Lecture 6:</i> Handling missing values and detecting outliers (Student Performance dataset). • <i>Lecture 22:</i> Interacting with a simple local intent-parser.

Assessment and Evaluation Framework

- **Marks break up**

• Attendance	5	
• Continuous Assessment	25	
• Mid-Term Test (All MCQ)	20	
• ETT (Mix: MCQ+Subjective)	50	

- **Total** 100

Details of Academic tasks

CA Category

A0203

Considering 2 Best out of Remaining 3 CAs; Equal Weightage to all CAs

Academic Task	Objective	Detail of Academic Task	Mapped Cos	Nature	Marks	Allotment / Submission Week
Project	Tests students ability to solve real world problems through innovative AI products or patentable solutions with focus on technical skills, creativity, and societal impact.	Students are required to create a LLM/Agentic Solution to solve a real-world problem statement. The Evaluation Rubrics of the Project will be: (a) Innovation/Benefit to Society and Prospective Outcomes (Publication, Patent, Revenue etc.): 10 marks, (b) Project Quality and Completeness in terms of Technical Skills – 10 marks, (c) Report + Presentation/Viva Q&A – 10 marks	CO1, CO2, CO3, CO4, CO5, CO6.	Group	30	Allotment: 3 rd week Submission: 11 th Week
Test	To evaluate learning of students based on their understanding and application of concepts learned.	Test will be based on subjective questions from the topics covered in Unit-1 and Unit-2 of the syllabus. The test may be a mix of (but not limited to) concept-based, numerical-based, coding-based questions. Every question can be of 5 or 10 marks.	CO1, CO2	Individual	30	Allotment: 4 th week Submission: 5 th Week
Recoup CA	To test the subject knowledge of the students and provide opportunity to improve their CA performance in case if low marks or any missed CA	Subjective Test covering syllabus from Units-1 through Unit-3.	CO1, CO2, CO3	Individual	30	Allotment: 11 th week Submission: 12 th Week

Learning Resources | OERs

Unit mapped	Broad topic	Sub Topic	Source Type	Source Title	*%age mapping (approx)	Source URL
Unit 1	Foundations of Artificial Intelligence	Introduction to AI, AI vs ML vs DL vs Data Science, Problem-solving using AI, Intelligent Agents, Agent Environment, Ethics in AI	NPTEL Course	Artificial Intelligence: Concepts and Techniques	90	https://nptel.ac.in/courses/106108705
Unit 2	Data Preparation and Machine Learning Workflow	Introduction to Numpy, Pandas, Matplotlib, Understanding data, Data Preprocessing, Feature Engineering, Train-Test, Cross Validation	NPTEL Course	Python for Data Science, IIT Madras	90	https://nptel.ac.in/courses/106106212
			Swayam Course Video Archives	A Basic Course in Machine Learning for All		https://learn.swayam2.ac.in/learn/imb24_mg126

Learning Resources | OERs

Unit mapped	Broad topic	Sub Topic	Source Type	Source Title	*%age mapping (approx)	Source URL
Unit 3	Supervised Machine Learning	Linear Regression, Decision Tree Regression, Logistic Regression, KNN, Decision Tree Classifier and Random Forest, Concepts of Bias, Variance, Underfitting and Overfitting	NPTEL Course	NOC: Machine Learning for Core Engineering Disciplines, IISc Bangalore	100	https://nptel.ac.in/courses/127108778
			Video Lecture	Paper P-15 (Machine Learning); Module M-11		https://epgp.inflibnet.ac.in/Home/ViewSubject?catid=fBYckQKJvP3a/8Vd3L08tQ==
Unit 4	Unsupervised Learning	Clustering (K-Means Clustering, Hierarchical Clustering, DBSCAN), Dimensionality Reduction and Principal Component Analysis	Reading Material (pdf)	Paper P-15 (Machine Learning); Modules M-23, M-24, M-27	90	https://epgp.inflibnet.ac.in/Home/ViewSubject?catid=fBYckQKJvP3a/8Vd3L08tQ==
			Swayam Course Video	A Basic Course in Machine Learning for All		https://learn.swayam2.ac.in/learn/imb24_mg126

Learning Resources | OERs

Unit mapped	Broad topic	Sub Topic	Source Type	Source Title	*%age mapping (approx)	Source URL
Unit 5	Model Evaluation and Improvement	Regression metrics, Cross-validation, Hyperparameter tuning, Grid Search, Random Search	NPTEL Course	NOC: Machine Learning for Core Engineering Disciplines, IISc Bangalore	90	https://nptel.ac.in/courses/127108778
		Confusion Matrix, Classification Metrics, Cross Validation	Swayam Course Video Archives	A Basic Course in Machine Learning for All		https://learn.swayam2.a.c.in/learn/imb24_mg126
Unit 6	AI for Intelligent Systems and Robotics	Computer Vision, NLP Fundamentals, Intelligent Systems and Robotics, Reinforcement learning, AI for Autonomous Robots, Future trends	NPTEL Course	NOC: Introduction to Application of Artificial Intelligence to Robotics, IIT Roorkee	60	https://nptel.ac.in/courses/106107001

Learning Resources | Relevant Websites

Sr.	Web address	Salient Features
RW-1	https://epgp.inflibnet.ac.in/Home/ViewSubject?catid=fBYckQKJvP3a/8Vd3L08tQ==	ePG-Pathshala Computer Science: Paper P-15 Machine Learning
RW-2	https://epgp.inflibnet.ac.in/Home/ViewSubject?catid=iLkSuZZ5a+koxhsE1m+YjQ==	ePG-Pathshala Information Technology: Paper P-06 Artificial Intelligence
RW-3	https://www.tutort.net/blogs/ai-vs-ml-vs-dl-vs-data-science-whats-the-difference	Data Science vs Artificial Intelligence vs Machine Learning

Learning Resources | **Audio-Visual Aids (AV)**

Sr.	Web address	Salient Features
AV-1	https://learn.swayam2.ac.in/learn/imb24_mg126	SWAYAM Course Archives: A Basic Course in Machine Learning for All, by Sumitra Padmanabhan, ATLAS SkillTech University, Mumbai
AV-2	https://learn.swayam2.ac.in/learn/ntr26_ed111	SWAYAM Course Archives: Fundamentals of AI, Dr. B. Surendiran & Dr. V. Vani, NIT Puducherry
AV-3	https://learn.swayam2.ac.in/learn/nou25_ma05	SWAYAM Course Archives: Generative AI for Everyday Life, Dr. Ridip Dev Choudhury and Dr. Nabankur Pathak, Krishna Kanta Handiqui State Open University, Guwahati
AV-4	http://www.ndl.gov.in/he_document/nptel/nptel/N_C_S_A_E_I_T_M_L_N_579657916NDLI	NDLI: Introduction to Machine Learning [NOC]
AV-5	http://www.ndl.gov.in/he_document/nptel/nptel/N_C_S_A_E_A_I_1649783701	NDLI: Artificial Intelligence
AV-6	http://www.ndl.gov.in/he_document/nptel/nptel/N_C_S_A_E_A_I_P_P_D_1986933744NDLI	NDLI: Artificial Intelligence(Prof.P.Dasgupta)
AV-7	http://www.ndl.gov.in/he_document/openstax_cnx/openstax/IN__O__1_I_t_P_P__598_	NDLI: Introduction to Python Programming, Chapter 15 Data Science (NumPy, Pandas, Visualization)

Placement & Career Relevance

Employability Skills
Developed

Industry
Applications

Target Job Roles

Higher Education &
Exams

Revenue Generation & Entrepreneurship Opportunities

Freelancing

Custom ML Development: Building bespoke regression or classification models for independent clients on platforms like Upwork or Toptal.

Data Preparation Services: Offering raw data cleaning, EDA, and feature engineering (using NumPy and Pandas) to businesses lacking data infrastructure.

Consulting

AI Workflow Integration: Advising local SMBs on deploying conversational AI and NLP chatbots for automated customer support.

Predictive Analytics Consulting: Utilizing clustering and association algorithms (K-Means, Apriori) to provide customer segmentation and market basket insights for retail and e-commerce.

Product Development & Start-ups

Micro-SaaS Solutions: Developing lightweight, scalable tools such as resume ATS optimizers or automated financial report generators utilizing Scikit-learn.

Niche AI Agents: Launching autonomous systems tailored for specific industries (e.g., appointment schedulers for clinics, legal clause summarizers).

Industry-Sponsored Projects

Smart Automation: Collaborating with manufacturing firms to deploy computer vision and object detection for automated assembly line quality control.

Sensor Data Processing: Partnering with tech companies to analyze IoT robotics data, applying dimensionality reduction (PCA) to optimize processing speeds.

Real-World Monetization

Digital Asset Sales: Packaging and selling optimized hyperparameter templates, prompt engineering libraries, or meticulously cleaned, domain-specific datasets.

Technical Content & Education: Monetizing specialized technical knowledge through ML engineering blogs, AI deployment tutorials, or premium newsletter subscriptions.

Top Industry Certifications

Certification / Course	Description	Relevant URL
AWS Certified Machine Learning Engineer (Associate/Specialty)	Validates technical skills in building, training, deploying, and maintaining machine learning workloads on AWS.	AWS ML Certification
Microsoft Certified: Azure AI Fundamentals	Covers foundational AI capabilities and demonstrates how to implement AI solutions using Microsoft Foundry.	Azure AI Fundamentals
Google Cloud Professional Machine Learning Engineer	Validates expertise in deploying cloud applications, monitoring operations, and managing enterprise ML solutions on Google Cloud.	Google Cloud Certifications
DeepLearning.AI TensorFlow Developer Certificate	A hands-on Coursera program focused on applying machine learning skills with TensorFlow to build and train powerful neural networks.	Coursera TensorFlow Certificate

Emerging Technologies and Industry Trends

Trend / Technology	Industry Impact & Relevance
Generative AI & LLMs	Moving beyond predictive analytics into autonomous content generation, code writing, and advanced conversational AI agents.
Explainable AI (XAI)	Rising demand for transparency and auditability to interpret how "black-box" AI models arrive at specific decisions for regulatory compliance.
MLOps (Machine Learning Operations)	Standardizing the lifecycle management, continuous integration, deployment, and monitoring of machine learning models in live production environments.
Edge AI & Autonomous Robotics	Processing machine learning algorithms locally on edge devices and sensors for low-latency, real-time automation without relying on constant cloud connectivity.



EDU REVOLUTION

Transforming Learning.
Empowering Futures.

UMS Navigation → LMS > Edu-Revolution: Be the Change

Why EDU-Revolution!!!

EDU REVOLUTION

Transforming Learning.
Empowering Futures.

Supports the 'Earn
Your Fee' vision
through **Revenue
Generation** activities

Enhance Real World
Skills ensuring
Industry readiness

Promote **Innovation
and Creativity** to
handle real world
challenges

Promote **Lifelong
Learning**

UMS Navigation → LMS > Edu-Revolution: Be the Change

Edu-Revolution interface: **UMS Navigation → LMS → Edu-Revolution: Be the Change**

EDU-Revolution Initiatives:



EDU-Revolution Benefits in CSE252

•RPL(NonCore) Options

- Students may acquire the required knowledge through relevant MOOC platforms such as *Coursera*, *NPTEL*, *Swayam*, or other similar platforms by completing a course of **at least 20 hours** related to *Artificial Intelligence or Machine Learning*. The syllabus of the selected MOOC course must be mapped with this course, with a **minimum overlap of 65% syllabus**. This will demonstrate the student's prior knowledge in the field and make them eligible to apply for RPL.

EduRev

- **How to Apply:**

- UMS link to apply for EDU Revolution initiatives:

UMS Navigation □ LMS □ EDU Revolution: BE THE CHANGE

- **Queries:**

- For **other queries** raise a ticket for resolution through **RMS** at the following path:

UMS Navigation □ Relationship Management System □ EDU Revolution: BE THE CHANGE

OR,

- Visit the Helpdesk at **Block 38-205B**

OR,

- Get your queries resolved through the **Helpline Number: 01824-520150.**



Feedback

- Scan this QR Code and leave your feedback regarding Lecture-0...





Any questions?

THANK

YOU!